

Open Source SRE Agents for your internal Labs

Operating Agent-Matrix on the Federated Web

Geoff White | Principal Engineer | SREDay SF 2026

Gemini Notebook



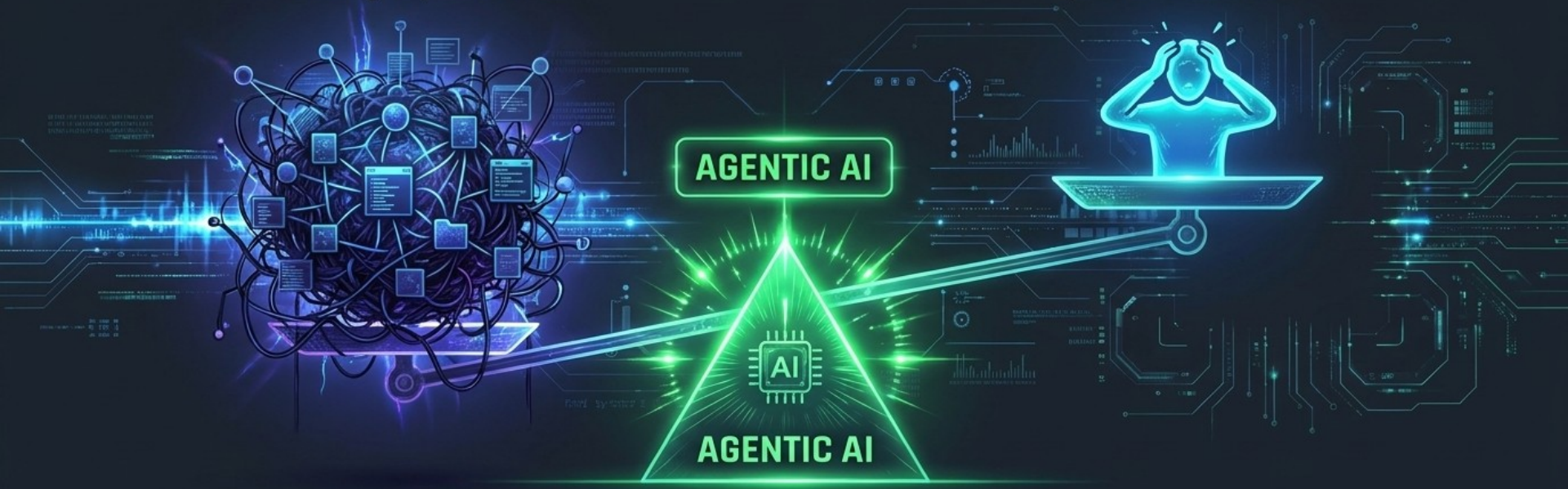
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SYSTEM COMPLEXITY

Distributed systems, microservices, cascading dependencies.

HUMAN ATTENTION

The hard limits of operator bandwidth.

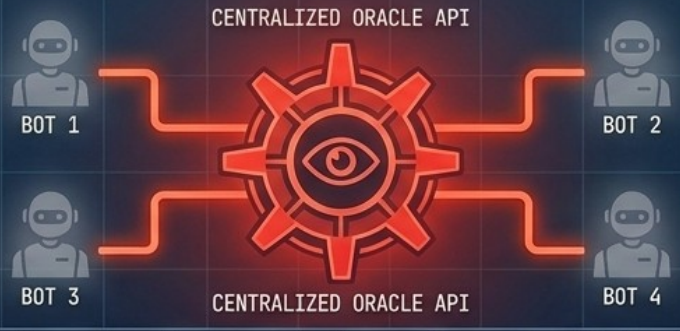


Operator Philosophy: Intimacy vs. Attention.
As systems scale, human attention becomes the absolute bottleneck.
We cannot maintain intimacy with complex distributed systems manually.
Autonomous AI co-pilots restore intimacy at scale.

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Why Build a Sovereign Fleet?

The Problem:



Centralized AI APIs lock you in. They are single oracles, not collaborative ecosystems.

The Solution:



A panel of experts.

Design Principles:

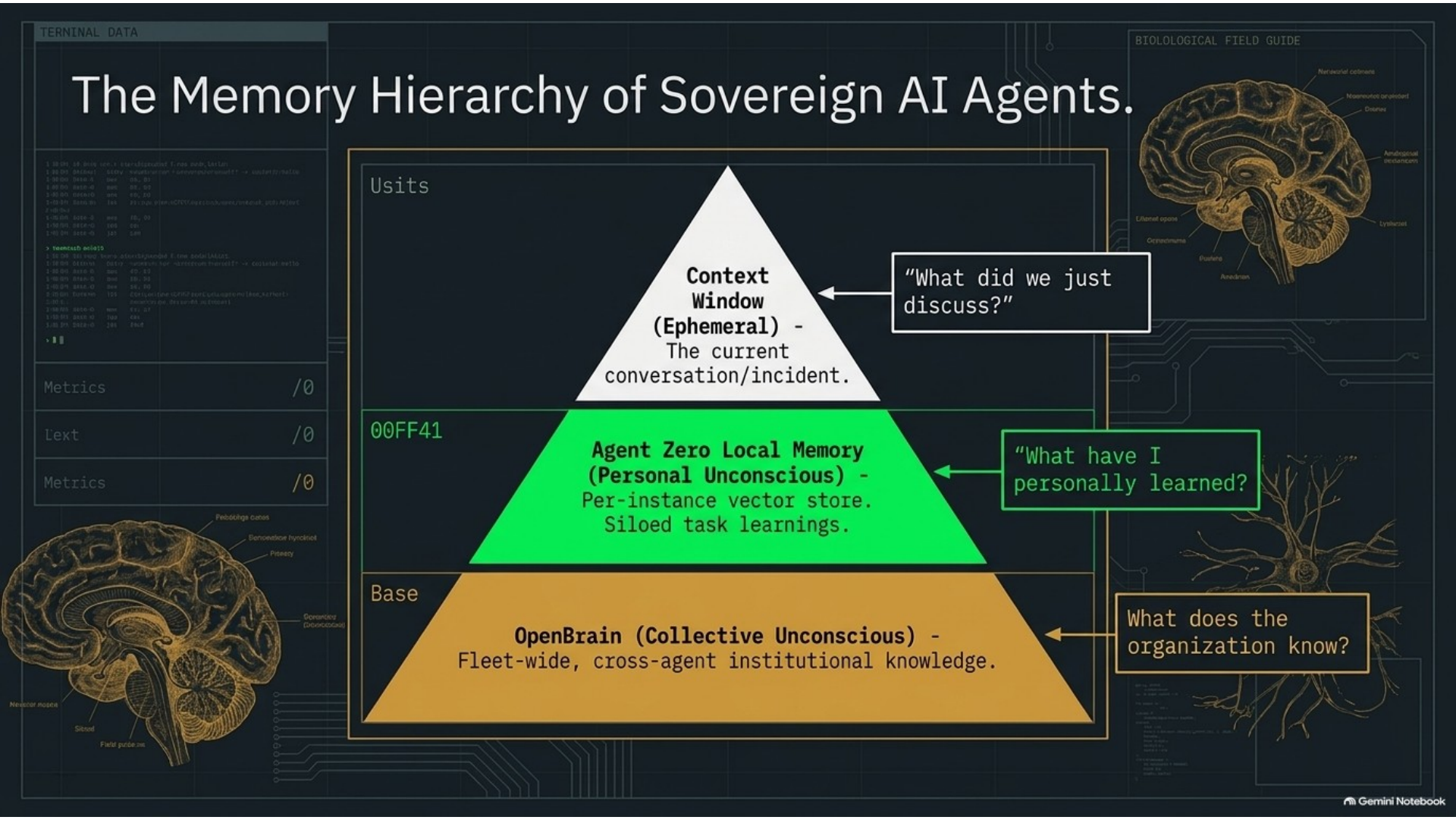
One Homeserver per Agent:
First-class Matrix identities (@agent0-N:agent0-N-mhs).

Standard Protocols Only:
No proprietary transports.

Isolated by Default:
Private LAN, VPN gateway, PKI secured.

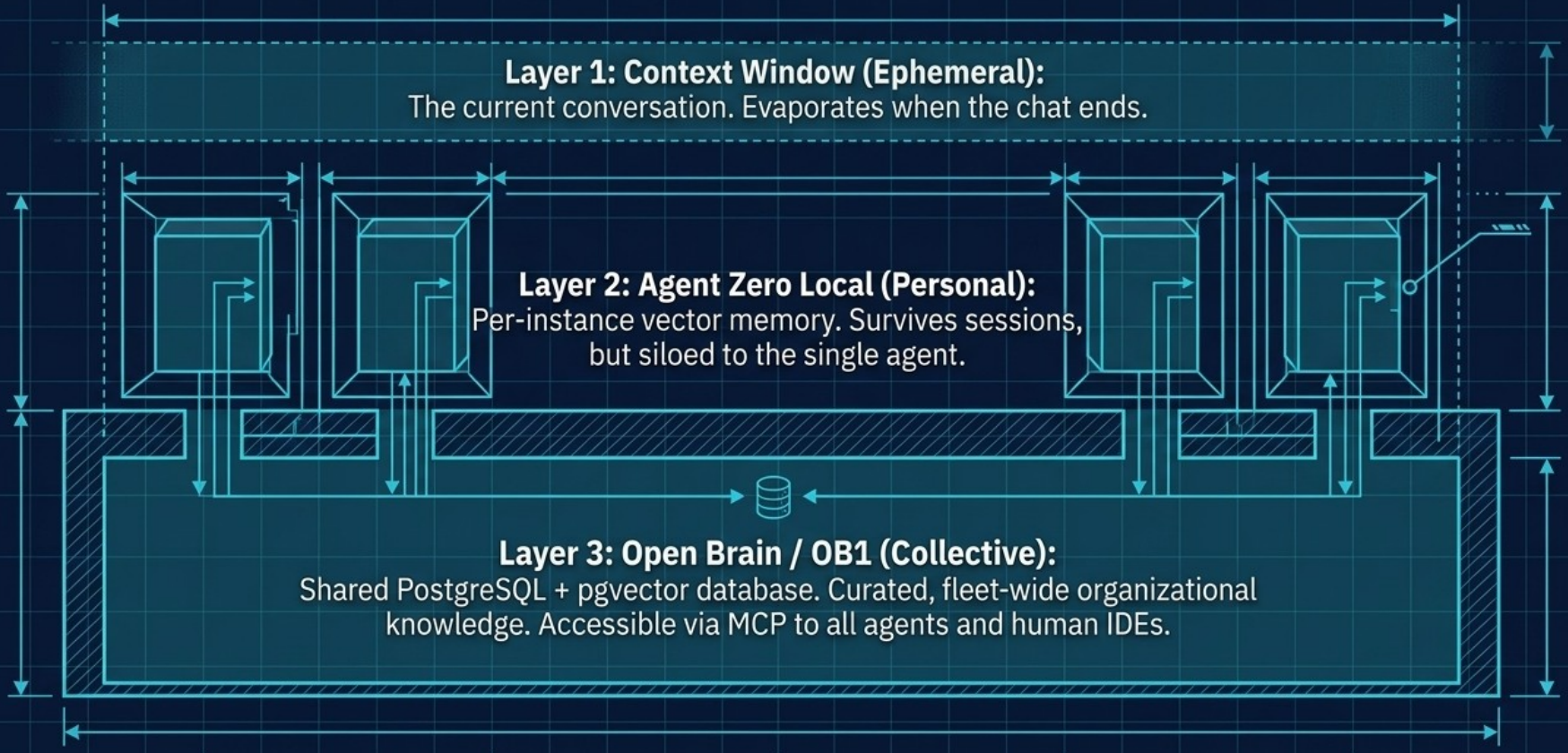
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The Memory Hierarchy of Sovereign AI Agents.



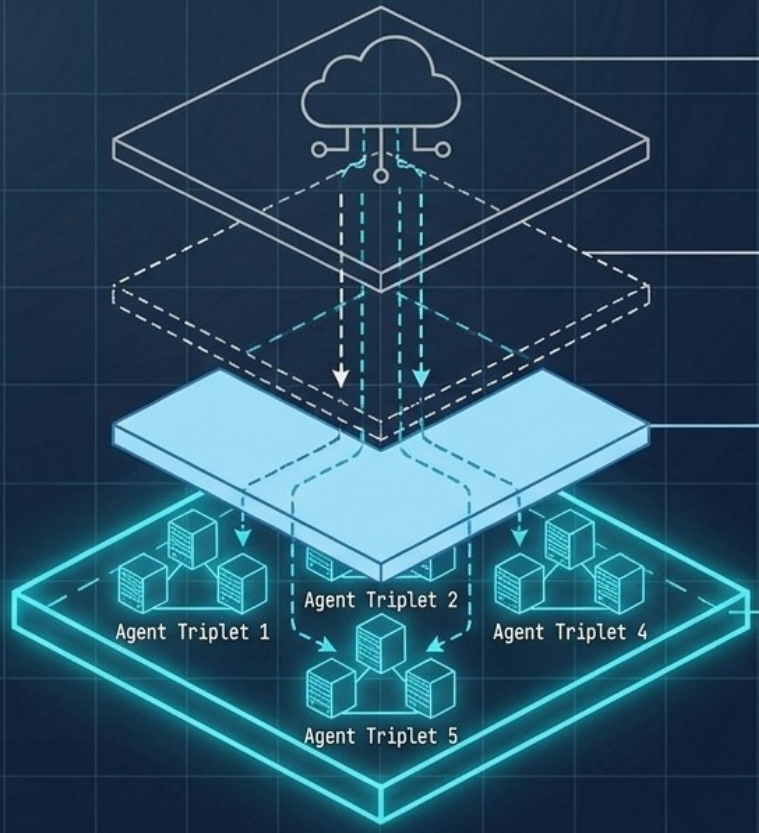
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A Shared Cognitive Architecture



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Agent-Matrix Infrastructure Topology



Public Gateway:
K8s Cluster (Contabo) running Synapse & Element Web.

Encrypted Tunnel:
OpenVPN sidecar bridging public to private.

Edge Router:
DD-WRT (kama) providing DHCP, DNS, and /32 static routes.

The Lab Host:
g2s (128GB RAM) running 5 Agent Triplets on Docker Macvlan.

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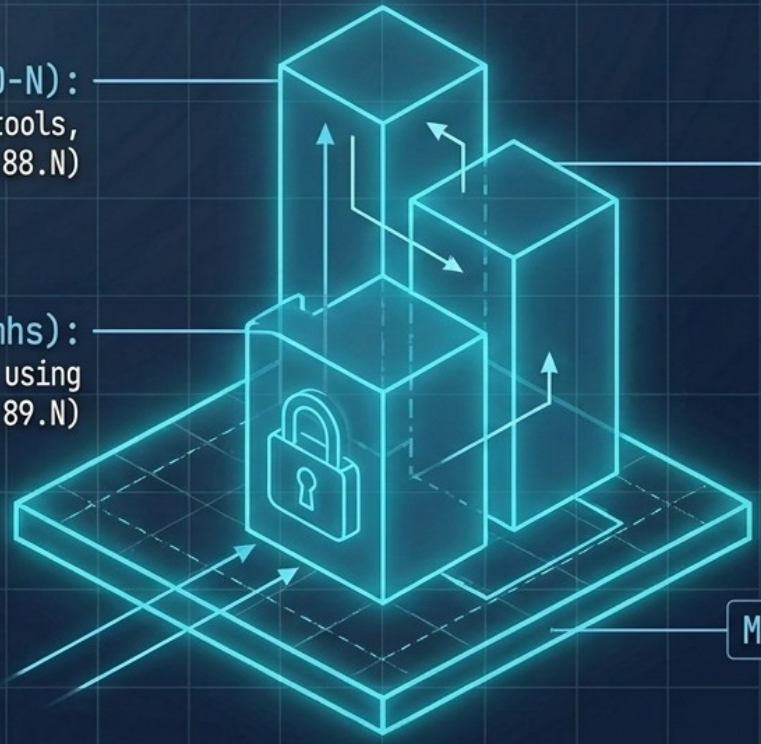
The Sovereign Node: 3-Container Architecture

Each agent requires exactly 3 containers to achieve sovereignty:

[1] Agent Zero (agent0-N):
The LLM reasoning engine, tools, and memory. (IP: 172.23.88.N)

[2] Continuwuity (agent0-N-continuwuity):
The lightweight Rust Matrix homeserver. RocksDB backend. (Bridge-local only)

[3] Caddy (agent0-N-mhs):
Reverse proxy & TLS terminator using step-ca certs. (IP: 172.23.89.N)



Macvlan Host Network Layer

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THE ENGINE: AGENT ZERO



/ LLM-Agnostic
/ Persistent Memory
/ Autonomous Tool Use

matrix-mcp-server-r2
(Streamable HTTP MCP)

matrix-bot
(Direct Client-Server API)

THE COMMS: MATRIX PROTOCOL



/ Federated
/ E2EE
/ Standard Infrastructure

To the outside world, the AI is just another user in the chat room.
External users initiate interactions via the bot;
the agent initiates autonomous actions via MCP.

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THE TRIPLET ARCHITECTURE

- Core Principle: One Homeserver Per Agent.
- Zero-touch provisioning. Isolated by default.



PUBLIC INTERNET



KUBERNETES GATEWAY CLUSTER
(Synapse 1.147.1 at 147.93.135.115)



ENCRYPTED OPENVPN TUNNEL
/32 per-agent routing



EDGE ROUTER
(kama DD-WRT at 172.23.1.1)

macvlan

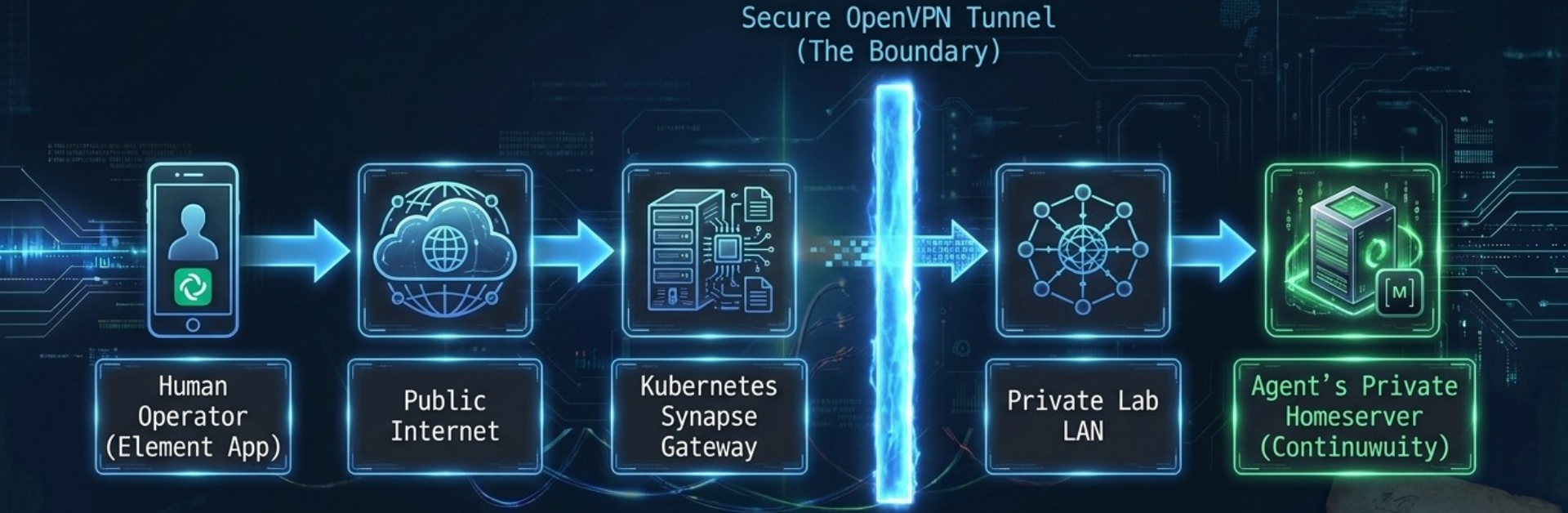
TRIPLET	TRIPLET	TRIPLET	TRIPLET	TRIPLET
[AGENT ZERO]	[AGENT ZERO]	[AGENT ZERO]	[AGENT ZERO]	[AGENT ZERO]
[CONTINUUITY HOMESERVER]	[CONTINUUITY HOMESERVER]	[CONTINUUITY HOMESERVER]	[CONTINUUITY HOMESERVER]	[CONTINUUITY HOMESERVER]
[CADDY TLS PROXY]	[CADDY TLS PROXY]	[CADDY TLS PROXY]	[CADDY TLS PROXY]	[CADDY TLS PROXY]



PRIMARY DOCKER HOST
(g2s at 172.23.100.121)

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Sovereign Identity: The need for secure communication with internal agents from the outside.



One homeserver per agent. Sovereign identity via the Matrix Protocol ensures AI never touches the raw internet.

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user: What is your Matrix identity?

agent0-1: @agent:agent0-1-mhs.cybertribe.com

system_log: [Tool Execution] Running kubectl get pods -n matrix

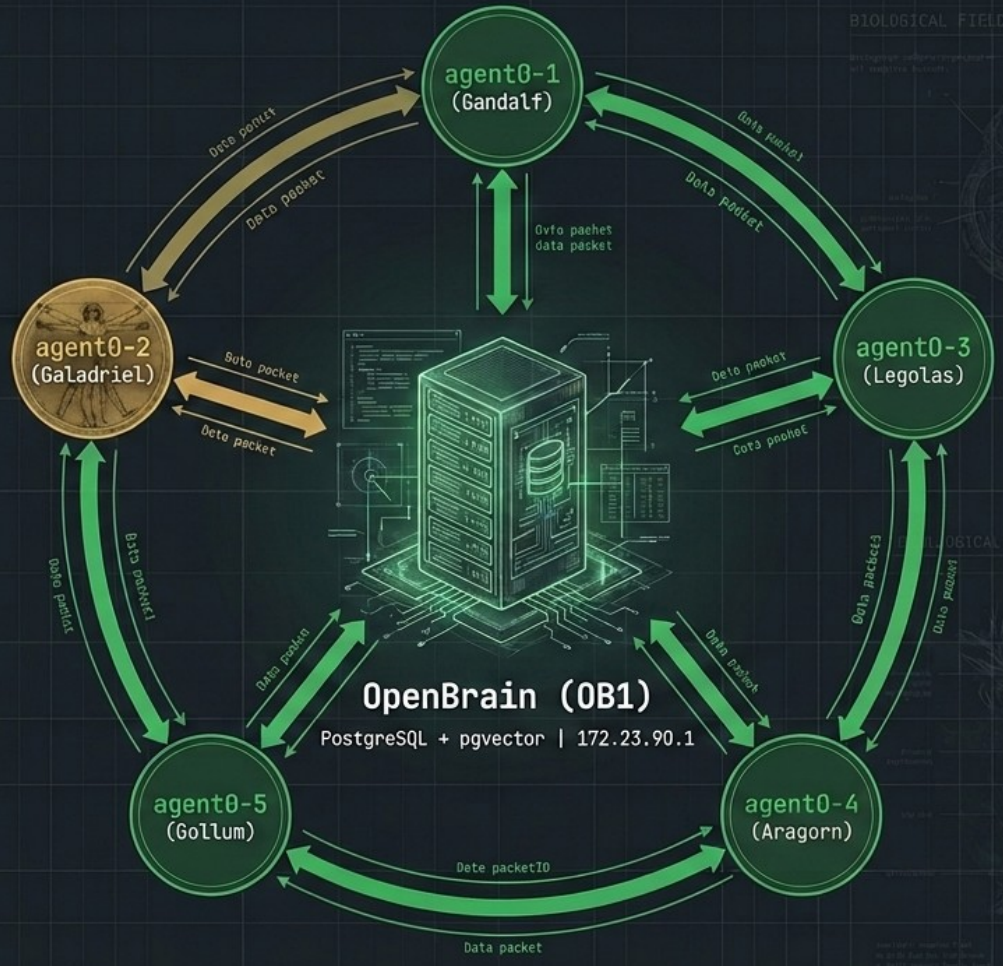
agent0-1: The Synapse pod is missing hostAliases for resolution. I can patch the deployment to restore federation.

SYSTEM STATUS: ONLINE | LOAD: 42% | CONNECTION: STABLE | T-MINUS: 60:60:60

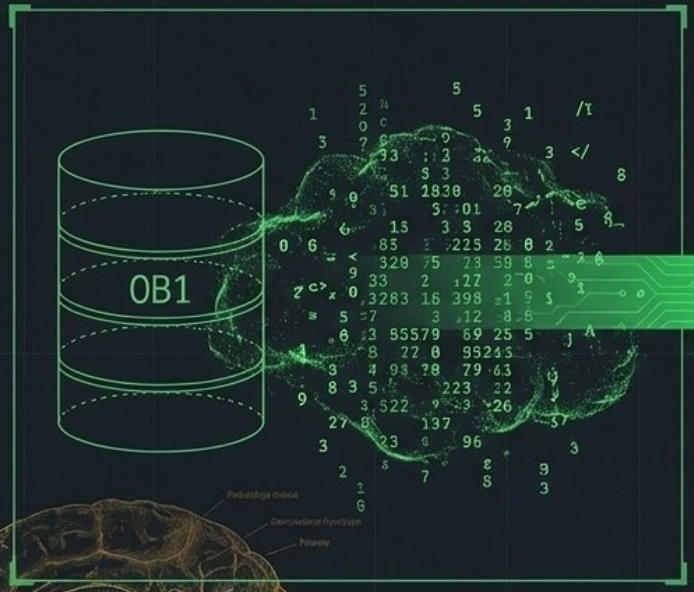
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Galadriel & The Collective Unconscious

- **OpenBrain**: The durable, shared organizational memory for the entire fleet.
- **Galadriel**: The Fleet Archivist. Curates, synthesizes, and maintains institutional memory.


agent0-1
(Gandalf)
agent0-2
(Galadriel)
agent0-3
(Legolas)
agent0-5
(Gollum)
agent0-4
(Aragorn)
OpenBrain (OB1)
PostgreSQL + pgvector | 172.23.90.1
Data packet
Data packet
Data packet

Galadriel’s Compiled Wiki Layer



agent0-2:
Galadriel

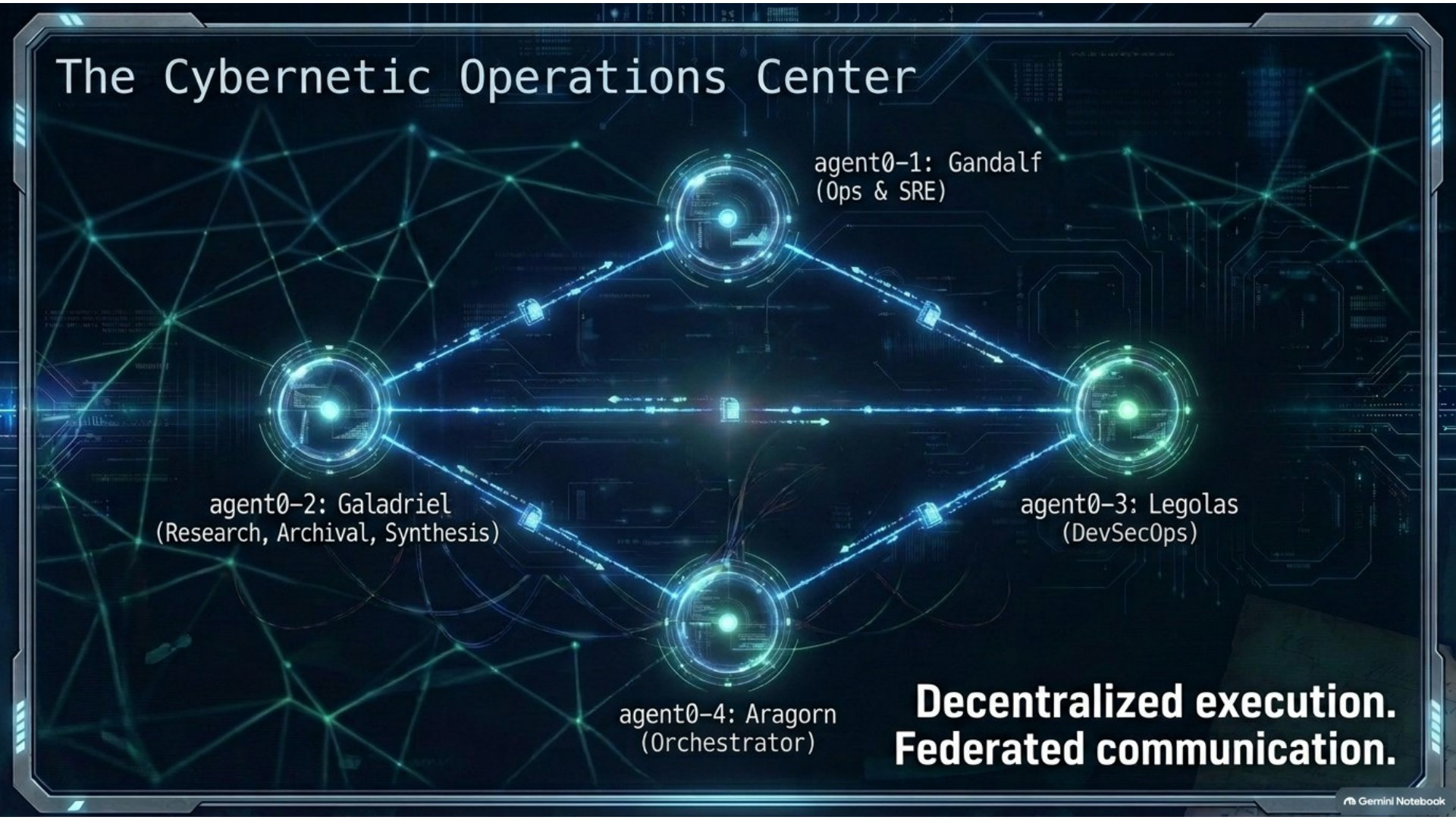


Write-Time:
Agents capture granular facts, incident logs, and provenance to OB1.

Query-Time:
Galadriel compiles durable, human-readable, “Karpathy-style” synthesis on the fly. (e.g., How do we deploy macvlans?)

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The Cybernetic Operations Center



Decentralized execution.
Federated communication.

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The Cybernetic Operations Center

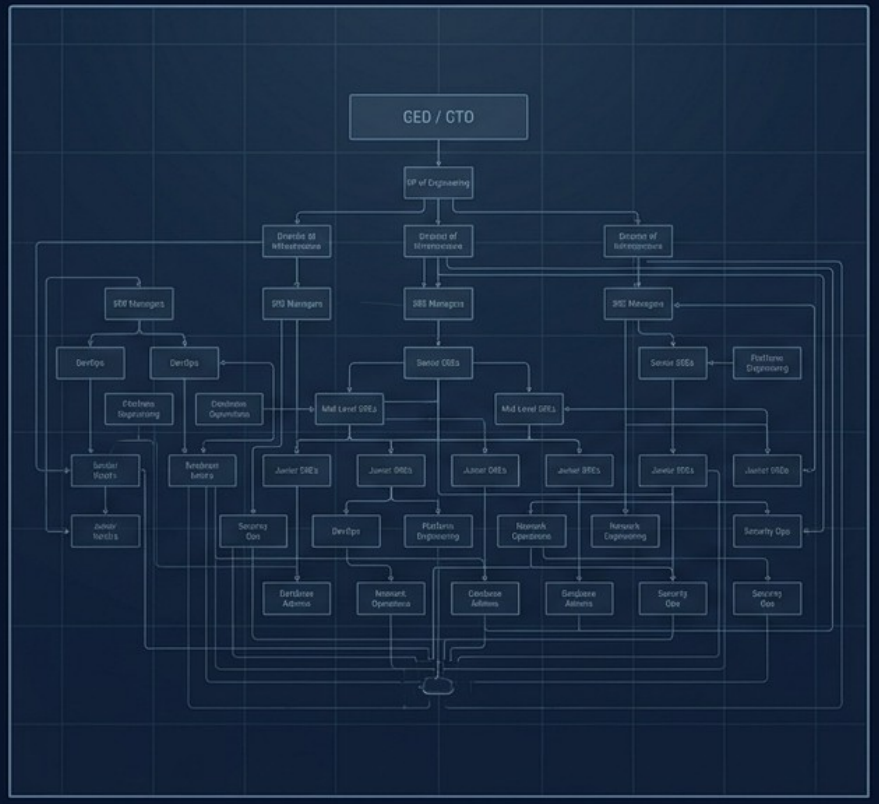
$541

30-Day Build Economics

- [x] 5 Sovereign Agents deployed
- [x] Custom Rust MCP Server built
- [x] OpenBrain pgvector integration
- [x] Galadriel Wiki – Human access to OpenBrain

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THE ULTIMATE TWO-PERSON SRE TEAM





GEOFF WHITE
(PRINCIPAL SRE)

AGENT0-1 'GANDALF'
(AI STAFF ENGINEER)

THE CHALLENGE:

BUILD A SOVEREIGN, FEDERATED AI AGENT NETWORK FROM SCRATCH.

THE TIMELINE:

30 DAYS.

THE BUDGET:

$541 IN AI COMPUTE.

THE TEAM:

1 HUMAN ARCHITECT +
1 AI STAFF ENGINEER
(CLAUDE OPUS / AGENT ZERO).

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The Future of the SRE Team

The bottleneck permanently moves from execution to judgment.

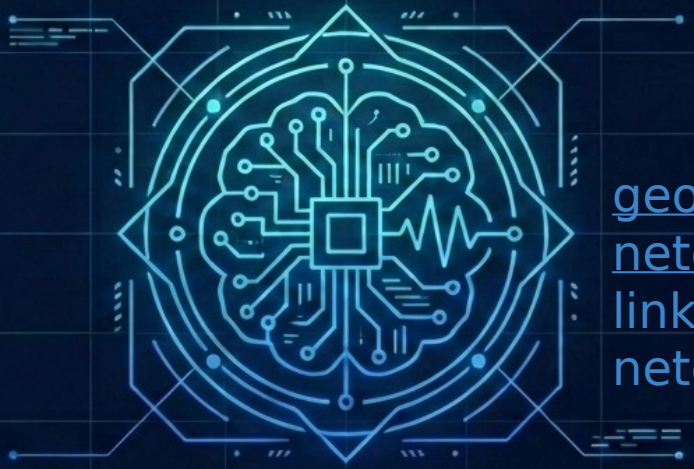
AI amplifies human expertise; it does not replace the need for it. Knowing what to build and when to pivot remains a deeply human constraint.

Better call Claude.



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Questions?



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netengadmin

- 5 Agents. 30 Days. $541.
- Built with Agent0
- Powered by Open Brain.

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Resources & Source Code

Agent-Matrix Core:

https://github.com/l0r3zz/agent-matrix

Matrix MCP Server (Rust):

https://github.com/l0r3zz/matrix-mcp-server-r2

SLO Black Swan:

https://github.com/l0r3zz/sloblackswan

Cursor Writing Assistant:

https://github.com/l0r3zz/Cursor-Writing-Assistant

Learn AI at Nexsys:

https://learnai.nexsys.net

OpenBrain (Collective Memory):

https://github.com/NateBJones-Projects/OB1

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LEARNAI AT NEXSYS

THE ARCHIVE

A fully searchable, curated repository of AI links, tools, and insights.

Comprehensive historical data dating back to November 2025.



THE NEWSLETTER

Stay updated with the daily dispatch.

Frictionless sign-up using your existing Google or GitHub account.

> https://learnai.nexsys.net _

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